HIKET — parameterization assumptions

Working document: what each parameter is, where its prior comes from, and a proposed consistent package

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Abstract

This document states every parameterization assumption in the HIKET calibration in one place, in the form they would take in a Methods section, and proposes a revised package. It was written after a diagnostic session (2026-08-07) on the first recalibration to include all round-1 changes and both upstream data corrections. That session found the assumptions to be individually defensible but *mutually inconsistent*: the same external evidence (the national growing-stock record) is used to set the *shape* of the pre-run litter ramp and simultaneously contradicted by the prior that sets its *level*; the two ends of the pre-run are referenced to different litter aggregates, so the parameter that controls the initial state does not mean what its name and documentation say; and the constraint introduced to keep litter fluxes physical bounds a quantity orthogonal to the failure it is now being asked to prevent. Sections 1–2 describe what is implemented; section 3 states the defects; section 4 proposes the replacement; section 5 derives the one new quantity it requires. A separate, unresolved problem — a uniform ∼20 % over-prediction shared by all six models — is recorded in section 7 because it conditions how the proposal should be read. A paired pre-flight (plain priors) additionally shows the bounded transform to be removable without cost, allowing the whole parameterization to collapse to four ordinary priors.

> **Naming.** Earlier drafts numbered the proposals P1–P5. They are now referred to by name, because the numbering repeatedly caused confusion with the model names (TP2/TP3) and with the C1–C5 revision series. The mapping, for cross-reference with older notes: P1 = COMMON ANCHOR; P2 = RATIO COORDINATE (now superseded); P3 = RATIO PRIOR; P4 = PLAIN PRIORS; P5 = FREE SLOW RATE.

1 The principle a prior has to satisfy here

The HIKET priors were built on a three-tier homogenization scheme whose stated aim is that priors be homogeneous in *construction method, scale, constraint and degrees of freedom* — not in numerical value. Tier 1 (climate, size) takes each model's centre from its own published calibration; Tier 2 (transfer fractions) applies a common weakly-informative logit prior; Tier 3 was the two auxiliary litter parameters, treated as nuisance quantities with an identical weak prior in every model.

That scheme has since been extended piecemeal. The auxiliary parameters acquired a physical window (the `flux_pair` transform, July 2026) and one of them acquired a physical centre (C4b), without a stated rule governing when a Tier-3 nuisance becomes a physical quantity. The result is the inconsistency this document addresses.

The rule we propose to adopt explicitly is:

> *A parameter with a physical referent takes its prior from external evidence about that referent, expressed in that referent's own units. A parameter without a physical referent should not be given an informative prior at all — and if it cannot be given one, that is a reason to reparameterize until the sampled quantity does have a referent.*

The second clause is the operative one. Both auxiliary parameters currently carry priors that are informative in practice, and one of them is a derived ratio whose prior cannot be stated in physical units at all.

A sharper operational version (added 2026-08-07). The rule above is about *referents*. In practice what determines how much a prior matters is the *curvature of the likelihood along it*, and the two auxiliary parameters sit in opposite regimes:

	likelihood cost of a 15 % move	prior cost	decided by
σ_{input}	69.3 nats	0.62 nats	**likelihood** ($\approx 112 : 1$)
σ_{init} / R	weak, and concentrated in the 1985 obs	—	**jointly** (see §8.5)

So: *a prior needs justification in proportion to how flat the likelihood is along it.* σ_{input} can carry a weak or even stale justification — the data overrules it by two orders of magnitude, which is why re-centring it from 1.30 to 1.00 leaves the posterior at 1.308 either way. R needs a defensible one, because the likelihood constrains it far more weakly — though *not* negligibly, see §8.5. This explains why the initialisation parameters have been the difficult ones throughout, and it means effort spent justifying C4b's σ_{input} centre is largely wasted while effort spent on R is not.

2 What is currently implemented

Everything in this section describes code that runs today — it is the parameterization behind the `20260805` calibration, not a proposal. The proposed changes are in section 4 and are clearly marked as such. Section 3 is the bridge: it states what is wrong with what follows here.

2.1 Decomposition rates

Yasso07/15/20 do not fit their decomposition rates: the a-vector is fixed at published values and injected as a constant, and what is calibrated is the twelve lateral transfer fractions, the climate parameters, woody size, and the two auxiliary σ's.

C1 (round 1) made the simple models homogeneous with that treatment. The fast rate is **hard-fixed** at the ICBM value $k_1 = 0.8$ with a one-time reference-climate offset ($\alpha_A = k_1/\xi_{\text{Ultuna}} = 0.851$); the slow rate stays free under a **very-informative** prior centred on k_2/ξ_{Ultuna}, on the argument that k_2 is the one weak link (it is measured on Ultuna *arable bare fallow*) and that the posterior width would then serve as an arable→forest transferability diagnostic. Prior SD was tightened from 0.50 to 0.15 on the log scale.

2.2 Humification and transfer fractions

Free in every model, under the common Tier-2 logit prior with SD 0.4, re-centred for the simple models at the ICBM humification coefficient $h = 0.13$. Fixing them would have made the simple models *more* constrained than Yasso, which the intercomparison design forbids.

2.3 The two auxiliary litter parameters

σ_{input} multiplies litter inputs; σ_{init} scales the litter flux at the start of the 1917 → 1985 pre-run.

Why they are not sampled directly. A prior placed on σ_{input} cannot express a physical belief, because σ_{input} is a bare multiplier with no referent — "$\sigma_{\text{input}} = 3$" is neither plausible nor implausible until you know what it multiplies. What *is* physically meaningful is the resulting litter **flux**, which has a boreal NPP envelope. So the `flux_pair` transform makes the two *fluxes* the sampled quantities and recovers the multipliers from them.

How the constraint works. The sampler proposes two ordinary unconstrained real numbers $x_1, x_2 \in (-\infty, \infty)$ — it never sees a bound and never rejects a proposal. Each is passed through the logistic, which maps the whole real line into $(0, 1)$, and the result is rescaled onto the physical window (a, b):

$$F \;=\; a + (b - a)\,\frac{1}{1 + e^{-x}} \;\in\; (a, b) \quad \text{for every } x \in \mathbb{R}. \tag{1}$$

The bound is therefore a property of the *coordinate system*, not a rule enforced during sampling: an out-of-envelope flux is not rejected, it is unreachable. As $x \to \pm\infty$ the flux approaches a or b asymptotically but never attains them. A Jacobian term $\log |\mathrm{d}F/\mathrm{d}x|$ is added to the posterior so that this stretching does not silently distort the prior.

The pair. Applying eq. (1) to both coordinates and recovering the model parameters:

$$F_{\text{now}} = a + (b - a)\,\text{logit}^{-1}(x_1), \qquad \sigma_{\text{input}} = F_{\text{now}}/\bar{J}, \tag{2}$$

$$F_{1917} = a + (b - a)\,\text{logit}^{-1}(x_2), \qquad \sigma_{\text{init}} = F_{1917}/F_{\text{now}}, \tag{3}$$

with $[a, b] = [0.05, 8.7]\,\text{tC}\,\text{ha}^{-1}\,\text{yr}^{-1}$ and $\bar{J} = 2.48\,\text{tC}\,\text{ha}^{-1}\,\text{yr}^{-1}$ the cross-plot mean litter, which converts an absolute flux into a dimensionless multiplier. It is called a *pair* because the two are coupled: σ_{init} is not sampled at all but emerges as the ratio of the two fluxes, and therefore depends on x_1 as well as x_2.

Worked example (Yasso20, current posterior). $\sigma_{\text{input}} = 1.388$ implies $F_{\text{now}} = 1.388 \times 2.48 = 3.44$; $\sigma_{\text{init}} = 1.236$ implies $F_{1917} = 1.236 \times 3.44 = 4.26$. Both sit inside $[0.05, 8.7]$, so nothing in the transform objects — even though the second says 1917 forests produced 24 % more litter than the contemporary flux. This is the concrete form of defect D3 below.

> **Two different ratios appear in this document and they are not the same number.** $\sigma_{\text{init}} = F_{1917}/F_{\text{now}} = 1.24$ compares the 1917 flux to the *contemporary* one. The quantity that decides whether the pre-run rises or falls is $J_{1917}/J_{1985} = \sigma_{\text{init}}/0.818 = 1.51$, which compares it to the *1985* flux. They differ because the two ends of the pre-run are referenced to different litter aggregates — which is defect D1, and the reason this parameter is so easy to misread.

Where the priors actually live. They are normal with SD 0.50 on the *unconstrained* coordinates x_1, x_2, centred at the values that map to $\sigma_{\text{input}} = 1.30$ (C4b, raised above 1 to stand in for the missing understorey component) and $\sigma_{\text{init}} = 1.0$. A consequence worth stating: because the prior sits on x rather than on the flux, its shape in physical units is not lognormal or anything else nameable — it is a logit-normal on the window, so its spread in $\text{tC}\,\text{ha}^{-1}\,\text{yr}^{-1}$ depends on where in the window it sits. This is why neither of us could say what the σ_{init} prior *was* until we read the transform code.

It fixed what it was built for. With a weak prior on the abstract multiplier, TP2 and TP3 had driven σ_{input} to 13–20×, implying effective litter fluxes of 34 and $50\,\text{tC}\,\text{ha}^{-1}\,\text{yr}^{-1}$ — above boreal NPP. After `flux_pair`, effective fluxes are 0.9–3.5 in all six models. That pathology is genuinely gone, which is also why the hard bound may no longer be load-bearing (section 4, decision 3).

2.4 Transient initialisation

Each plot is started at steady state under a 1917 flux and the flux is then ramped to its 1985 value over 68 years:

$$J_{1917} = \bar{J}_{\text{full}}\,\sigma_{\text{init}}\,\sigma_{\text{input}}, \qquad J_{1985} = \bar{J}_{t_0}\,\sigma_{\text{input}}, \tag{4}$$

where $\bar{J}_{\text{full}}$ is the plot's mean litter over the *whole* 1986–2024 series and $\bar{J}_{t_0}$ its mean over the *first five* years. The ramp between them follows the national growing-stock trajectory (C3, Korhonen et al. 2024), which replaced a linear ramp; the source file states explicitly that "litter scales with standing biomass" and equally explicitly that "the calibrated endpoints J_{1917}/J_{1985} are UNCHANGED; only the shape between moves."

2.5 Likelihood

Multiplicative normal, with the SD proportional to the prediction and its coefficient fixed:

$$\text{SOC}^{\text{obs}} \sim \mathcal{N}\left(\widehat{\text{SOC}},\ \left(\widehat{\text{SOC}} \cdot \sigma_{\text{obs}}\right)^2\right), \qquad \sigma_{\text{obs}} = 0.4416. \tag{5}$$

σ_{obs} is estimated externally from the observation campaigns and held fixed. There is no separate model-error term.

3 Defects identified 2026-08-07

DEFECT — D1 — the two ends of the pre-run use different litter aggregates

In eq. (4) the 1917 flux is referenced to $\bar{J}_{\text{full}}$ (the 1986–2024 mean) and the 1985 flux to $\bar{J}_{t_0}$ (the 1986–1990 mean). Since litter rises through the record, $\bar{J}_{t_0}/\bar{J}_{\text{full}}$ has median 0.818 over the 512 plots. Therefore

$$\frac{J_{1917}}{J_{1985}} = \sigma_{\text{init}} \cdot \frac{\bar{J}_{\text{full}}}{\bar{J}_{t_0}} = \frac{\sigma_{\text{init}}}{0.818},$$

so σ_{init} is *not* the 1917/1985 flux ratio: it is that ratio scaled by an arbitrary choice of reference aggregate. $\sigma_{\text{init}} = 1$ does not mean "1917 like 1985"; it means "1917 flux 22 % above 1985". `CLAUDE.md` additionally still describes σ_{init} as a "proportional SD on the initial carbon state (log-transformed)", which was true before `flux_pair` and is now wrong in both respects.

DEFECT — D2 — the initialisation prior is informative in the wrong direction

The default $\sigma_{\text{init}} = 1.0$ therefore asserts a 1917 litter flux 22 % *above* 1985. The growing-stock record — the same source used for the ramp shape — implies a ratio below one (section 5). The prior centre is roughly 1.5× too high on the quantity that sets the initial carbon state. This is not a weak prior that the data can overrule: the posterior σ_{init} moves $+0.00\sigma$ (Yasso15) to $+0.42\sigma$ (Yasso20) from that centre.

DEFECT — D3 — the constraint bounds levels, not the ratio

`flux_pair` bounds F_{now} and F_{1917} each to $[a, b]$, but the pre-run *direction* depends on their ratio, which can take any value in $[a/b,\ b/a] = [0.006,\ 174]$ without approaching a boundary. In the current run Yasso20 has $F_{1917} = 4.26$ against $F_{\text{now}} = 3.44$ — more litter in 1917 than today — with both fluxes comfortably inside the envelope. The constraint is orthogonal to the failure.

Consequences in the current run: the pre-run declines for 78 % (Yasso15) and 92 % (Yasso20) of plots, those two models over-predict the 1985 stock by $+19.0$ and $+23.5\,\text{tC ha}^{-1}$, and both report a soil carbon *source* (-0.063, $-0.155\,\text{tC ha}^{-1}\,\text{yr}^{-1}$) where the three campaigns show a *sink* ($+0.312$). The sign error survives dropping the 1985 campaign entirely.

DEFECT — D4 — the likelihood conflates observation error with model error
σ_{obs} in eq. (5) is fixed at the observation CV, and no model-error term exists. Actual residuals exceed the assumed SD by a factor 1.19–1.21 in *all six* models. Because the SD is proportional to the prediction, the only way the fit can widen its error bars is to raise the predictions.

4 Proposed package — NOT IMPLEMENTED

Nothing in this section is in the code. Section 2 is what runs today; this is what we propose to replace it with. the common anchor is a defect correction we would make regardless; the ratio coordinate and the ratio prior are conditional on the unresolved level offset (section 7) — see section 6 for why they make two models visibly worse until that is understood.

The three proposals below are stated as a package because they are not independent: the ratio coordinate is what makes the ratio prior expressible, and the common anchor is what makes the ratio coordinate mean what it says.

PROPOSAL — COMMON ANCHOR — reference both ends of the pre-run to the same aggregate
Replace eq. (4) with

$$J_{1985} = \bar{J}_{t_0}\,\sigma_{\text{input}}, \qquad J_{1917} = \bar{J}_{t_0}\,\sigma_{\text{input}}\,R, \tag{6}$$

so that R *is* the 1917/1985 litter ratio. $\bar{J}_{\text{full}}$ ceases to be used as a historical anchor, which it should never have been: it is a modern-record mean standing in for a historical quantity. σ_{init} is retired and replaced by R.

Justification. This is a defect correction, not a modelling choice. It should be made regardless of what is decided about priors or bounds.

PROPOSAL — RATIO COORDINATE — make the second `flux_pair` coordinate the ratio
Keep the NPP envelope on the contemporary flux; sample the ratio rather than the 1917 flux:

$$\begin{aligned} F_{\text{now}} &= a + (b-a)\,\text{logit}^{-1}(x_1), & \sigma_{\text{input}} &= F_{\text{now}}/\bar{J}, \\ R &= R_{\text{lo}} + (R_{\text{hi}} - R_{\text{lo}})\,\text{logit}^{-1}(x_2), & J_{1917} &= \bar{J}_{t_0}\,\sigma_{\text{input}}\,R, \end{aligned}$$

with $[R_{\text{lo}}, R_{\text{hi}}] \approx [0.25, 1.25]$. The Jacobian gains a bounded term for R and loses the $-\log F_{\text{now}}$ term, since the ratio no longer depends on the contemporary flux. A guard returns $-\infty$ if J_{1917} leaves the NPP envelope; with $F_{\text{now}} \approx 3.2$ this is far from binding, but it preserves the physical guarantee explicitly rather than by assumption.

Justification. The pre-run direction is the quantity that fails, and it is the quantity for which external evidence exists. Placing the prior on it is the only way to state that prior in physical units.

PROPOSAL — RATIO PRIOR — derive the prior on R from the growing-stock record
$R \sim \text{Lognormal}(\log 0.90,\ 0.15\text{–}0.20)$, derived in section 5.

Justification. It uses the same NFI source, and the same interpolation, that already sets the ramp shape — removing the inconsistency in which the pipeline invokes "litter scales with standing biomass" for the shape and abandons it for the level.

This prior will do real work, and that must be stated. The likelihood is nearly flat in this direction: a 20 % error in the 1985 stock is half of σ_{obs}, and the C5 sweep moved σ_{init} by a factor of four while changing trusted-campaign RMSE by 0.45 %. A prior that dominates a direction the data cannot see is not a neutral choice.

The defence is not that the prior is weak, but that *the status quo also dominates that direction* — asserting $R \approx 1.22$ — and does so in contradiction with evidence the same pipeline already uses. Replacing an unjustified informative prior with a justified one removes an inconsistency rather than adding an assumption. The sensitivity should nevertheless be reported, as the C5 sensitivity now is.

plain priors — can `flux_pair` simply be removed?

the common anchor to the ratio prior keep the bounded transform and change what the second coordinate means. A simpler question is whether the transform is needed at all, since its cost is precisely that its priors cannot be stated in physical units (section 2).

Why it might not be. `flux_pair` was introduced against a genuine pathology: under the old Tier-3 prior (log SD 0.50) TP2/TP3 drove σ_{input} to 13–20×. But an SD-0.50 lognormal places ≈ 2.4 % of its mass above the NPP ceiling, whereas an SD-0.30 lognormal places ≈ 0 %. The containment may therefore be achievable with a tighter *ordinary* prior. Indeed the prior currently implied by the transform is nearly indistinguishable from Lognormal(log 1.30, 0.30): quartiles 1.04/1.30/1.58 against 1.06/1.30/1.59, differing only in a slightly thinner right tail (2.14 vs 2.34 at 97.5 %).

Test. `doublechecks/preflight_naive_priors.R` patches the real calibration script to swap `flux_pair` for two `log` parameters at SD = 0.30 / 0.20, leaving everything else — anchoring, centres, data — untouched, and pushes $K = 200$ prior draws through the genuine `ll_fn` at full N. Both configurations use the same seed, so the comparison is paired.

Table 1: Prior-pushforward, $K = 200$ draws at full $N = 410$. "rejects" are legitimate stick-breaking simplex rejections; "blow-ups" are non-finite ℓ among constraint-passing draws — the MCMC `inf_rate` proxy.

model	naive (no `flux_pair`)	production (`flux_pair`)	rejects (both)
SP1	0 / 200	0 / 200	0
TP2	0 / 200	0 / 200	0
TP3	0 / 200	0 / 200	0
Yasso07	5 / 101 (5.0 %)	5 / 101 (5.0 %)	99
Yasso15	0 / 145	0 / 145	55
Yasso20	0 / 145	0 / 145	55

Identical in every model. Yasso07's five blow-ups occur in both configurations among the same 101 constraint-passing draws, so they are the known Yasso07 δ_2/σ baseline rather than a consequence of removing the bound. Implied effective flux under the naive prior is median 2.97–3.35 with 97.5 % at 4.94–6.34 $\text{tC ha}^{-1}\,\text{yr}^{-1}$, and 0 % **of prior mass exceeds the NPP ceiling in all six** — the bound would have had nothing to remove.

PROPOSAL — PLAIN PRIORS — the simplified parameterization

$$\sigma_{\text{input}} \sim \text{Lognormal}(\log m,\ 0.30) \qquad m\text{: see decision 5}$$
$$R \sim \text{Lognormal}(\log 0.90,\ 0.20) \qquad \text{growing-stock elasticity, §5}$$
$$J_{1985} = \bar{J}_{t_0}\,\sigma_{\text{input}}, \qquad J_{1917} = \bar{J}_{t_0}\,\sigma_{\text{input}}\,R$$
$$\text{guard : reject if the effective flux exceeds the NPP ceiling}$$

Four statements, each expressible in one sentence of Methods, each carrying a prior on a quantity with a referent. It subsumes the common anchor to the ratio prior and *dissolves* D1 and D3 rather than repairing them: there is no level-versus-ratio mismatch left to have.

This tests the prior, not the posterior. The pre-flight confirms the forward model stays finite across prior mass. It cannot rule out the sampler wandering into a large-σ_{input} region that the bound would have blocked — a lognormal has an infinite tail where the logistic has a wall, and the 13–20× excursions were a *posterior* phenomenon. Only a real calibration settles that, which is why the guard is retained as a truncation that should never fire. Removing `flux_pair` would also require reframing the F9 result, currently stated as "the flux bound closed the σ_{input} escape hatch"; the honest replacement is that the escape hatch was a symptom of an inflated SOC target and a starved cascade, and that once those were corrected an informative physical prior sufficed.

5 Deriving R

Why not the volume ratio. The NFI record gives growing stock 1400 Mm^3 at 1917 (NFI1, 1922, held flat back over a near-stationary period) and 1775 Mm^3 at 1985, a ratio of 0.789. Taking $R = 0.789$ assumes litter is *proportional* to standing volume. Three arguments say it is not: foliage and fine-root litter scale sub-linearly with stem volume because leaf area saturates as stands close; woody litter tracks mortality rather than standing stock, and the mortality regime of cut-over unmanaged forest differs; and understorey litter — absent from the litter product entirely — is proportionally *larger* in sparse stands. All three bias R upward relative to the volume ratio.

Empirical elasticity. Rather than assume an exponent, it can be fitted. Both series exist over 1986–2023: national mean litter from the input bundle, and growing stock interpolated from the same NFI file used for the ramp shape. Fitting $J \propto \text{GS}^{\epsilon}$ and extrapolating $R = (\text{GS}_{1917}/\text{GS}_{1985})^{\epsilon}$:

Table 2: Litter–growing-stock elasticity over 1986–2023 (growing stock +42 %, mean litter +35 % over the window), and the implied 1917/1985 litter ratio.

fit	ϵ	$R = 0.789^{\epsilon}$	R^2
annual, all years	0.467	0.895	0.370
5-year block means	0.448	0.899	0.401
endpoints only (5-yr means)	0.661	0.855	—
naive proportionality	*1.000*	*0.789*	—

The elasticity is close to 0.5 — litter scales roughly with the square root of growing stock — and is stable against interannual climate noise, which the low R^2 of the annual fit makes the obvious concern. The regression SE on ϵ (0.101) alone would give $R \in [0.854, 0.938]$.

Adopted value. We propose centring at $R = 0.90$ with log SD 0.15–0.20, wider than the regression SE alone would justify, to accommodate the caveats below.

> **Five caveats on this derivation.** (i) *Possible circularity* — if the litter product is itself derived from biomass, this elasticity is internal to the product rather than independent evidence. For the purpose of extrapolating *that product* backwards consistently it is arguably the correct choice, but it requires confirmation from B. Tupek on how the product is constructed. This is the caveat most likely to matter to a reviewer. (ii) *Extrapolation* — the relation is fitted over GS 1798–2552 and applied at 1400, 22 % below the fitted range. (iii) *Understorey omitted* — the product is tree litter only, so the total-litter ratio is biased *upward* from this estimate, plausibly 0.92–0.95. Our value is a lower bound. (iv) *A national ratio applied per plot* — individual plot histories differ from the national mean. (v) *Non-stationarity* — management, species composition and age structure changed over 1917–1985, so the modern volume-to-litter relation need not have held historically.

6 What the package fixes, and what it does not

Fixes. σ_{init} is retired in favour of a parameter with a referent and a stateable prior; the pre-run direction becomes evidence-constrained; the shape/level inconsistency in the C3 implementation disappears; and — if the mechanism holds — the Yasso15/20 pre-run inversion and their inverted sink go with it.

It does not fix the level. A uniform over-prediction of 17–20 % is shared by all six models (section 7), and the common anchor to the ratio prior do not address it.

And it makes two models visibly worse. At $R = 0.90$ the equivalent σ_{init} is ≈ 0.74. Against the current posteriors:

Table 3: Current σ_{init} against the proposed centre (displacement at log SD 0.20).

model	σ_{init}	vs. proposed centre
SP1	0.627	-0.8σ
TP2	0.564	-1.4σ *(too low)*
TP3	0.566	-1.4σ *(too low)*
Yasso07	0.768	$+0.2\sigma$
Yasso15	1.002	$+1.5\sigma$
Yasso20	1.236	$+2.6\sigma$ *(too high)*

The constraint pulls Yasso15/20 down, which is the intended repair, but pulls TP2/TP3 *up*, giving them more 1917 input and worsening their level over-prediction. TP2 and TP3 have evidently been suppressing their 1917 input below what the forest history supports, partially offsetting the level bias. Pinning R removes that compensation and forces the misfit into the open — the same logic by which the `flux_pair` bound became a result rather than a cosmetic fix. It does, however, mean the two problems cannot be adjudicated independently: if R is pinned before the level offset is understood, the simple models will get visibly worse and it will not be clear whether that is the constraint working as intended or a second error compounding.

7 Open: the uniform level offset

All six models over-predict by 17–20 % (median log-residual −0.176 to −0.206; campaign means +22–38 % at 1985 falling to +7–14 % at 2024). Three explanations were tested on 2026-08-07 and none is fully satisfactory.

- *Not the input data.* σ_{input} is a free per-model multiplier on exactly that quantity and demonstrably moves (SP1 sits 2.6σ from its prior centre; no flux bound binds). Any uniform input bias is absorbed by construction.

- *Not the σ_{input} prior centre.* SP1 is 2.6σ *below* the C4b centre and still over-predicts by 24 %; and the excess is 23–28 % in all six models while σ_{input} itself spans 0.35–1.39.

- *Not insufficient kinetic freedom.* The rates moved 1.6–2.2σ against a prior of SD 0.15 — but *toward longer* residence times (+28 to +38 %), i.e. *raising* SOC. Both available routes to lowering the level were left unused or used in the opposite direction.

- *Partly the likelihood form.* Under eq. (5) the optimal uniform rescaling of the predictions is $k = 1.019$: the likelihood positively prefers the current over-prediction. Freeing σ_{obs} within the same form moves this only to $k = 0.960$; changing the form to log-normal moves it to $k = 0.848$, against 0.811 for matching the observed median. So the *form* accounts for most of the offset and the *width* for little — but this is inference from fitted predictions, not a refit.

> The decisive test has not been run: refit one model (TP2) under a log-normal likelihood with nothing else changed. If the offset collapses and σ_{input} lands near the 1.04 the observations require, the fitting criterion is responsible. If it does not, the cause is neither the inputs, the priors, the kinetics nor the likelihood, and remains unidentified.

A note on the rate displacement. C1's design states that the slow-rate posterior width would serve as an arable→forest transferability diagnostic. That diagnostic has fired: all three simple models sit 1.6–2.2σ from the ICBM anchor, consistently slower, implying Finnish forest slow-pool turnover some 30 % slower than the Ultuna arable value. Whether this is a genuine transferability result or the rates absorbing misfit that belongs elsewhere is testable — if the latter, correcting the level should pull them back toward the anchor.

8 The level and the trend are different problems

Section 7 recorded a uniform ∼20 % over-prediction of unknown cause. It has since been identified, tested and largely removed — and in doing so it separated cleanly from a second problem that it had been masking.

8.1 The level: an error-model artefact, now fixed

The multiplicative normal of eq. (5) sets the variance from the parameter being estimated, so a prediction widens its own error bar. The penalty for a badly-missed plot therefore *plateaus* rather than growing, and the fit buys tolerance for its worst plots by inflating everything. The consequence is a location estimate contaminated by dispersion. On data generated *unbiased* by construction, the optimal scaling under this form drifts from 0.86 to 3.25 as the residual CV rises from 0.05 to 1.0, while a log-normal returns 1.000 at every spread.

A refit of TP2 with the likelihood replaced by a log-normal (3 chains × 6000, C5 = 1.0; control is the matched `A1_C5_off` ablation) confirms it:

Table 4: TP2 under the two error models, at posterior-median parameters.

	multiplicative normal	log-normal
bias 1985 ($\mathrm{tC\,ha^{-1}}$)	+15.2	+2.8
bias 2006	+11.6	−2.2
bias 2024	+11.5	−3.1
bias overall	**+12.7**	**−0.9**
median log-residual	−0.205	−0.023
paired trend 1985–2024	+0.168	+0.116
observed	+0.309	

The bias collapses to essentially zero. The implied rescaling, 0.834, matches the 0.848 predicted from the closed-form analysis to within 2 %. Convergence is acceptable (R-hat 1.009–1.033, ESS 206–382). **The level bias was ours, not the models'** — which means every level-based number from the `20260805` run carries it.

> The route was not the one predicted. σ_{input} barely moved (1.290 → 1.306); the model took the level down through the climate parameters instead (β_1 +35 %). The *magnitude* of a refit is predictable from a fixed-prediction rescaling analysis; the *parameter that delivers it* is not. Relatedly, σ_{init} rose (0.563 → 0.622, R 0.69 → 0.76), so part of the apparent "the data already infers historical depletion" signal was the biased likelihood.

8.2 The trend: a response-time problem, with two distinct causes

Correcting the level did not fix the trend — it slightly worsened it, and changed its signature from a uniform offset to the model *crossing* the observations. That is expected: a trend is governed by how far below its own equilibrium a model starts, and that ratio is invariant under uniform rescaling.

Fitting an exponential approach $C(t) = E - (E - C_0)e^{-t/\tau}$ to each model's own trajectory (1985 / 2024 / 2084) gives its effective response time:

Table 5: Effective response time and starting position, fitted per model. The observations require $\tau \approx$ 44–87 yr and a start at ≈ 0.73 of equilibrium.

model	τ (yr)	start/eq	gap closed in 39 yr	2024 if re-initialised at 61.5	
SP1	79	0.79	39 %	75.4	(+0.357)
TP2	**195**	0.68	**18 %**	70.8	(+0.237)
TP3	**194**	0.68	**18 %**	70.9	(+0.240)
Yasso07	38	0.93	64 %	76.3	(+0.381)
Yasso15	**17**	**1.03**	90 %	77.6	(+0.412)
Yasso20	23	**1.08**	82 %	76.3	(+0.379)
observed	*44–87*	*0.73*	*36–57 %*	*75.4*	*(+0.309)*

The failure has **two different causes**:

Yasso (and SP1): an initialisation problem. Their response times are already at or beyond the required range. All three Yasso models start at or *above* their own equilibrium (start/eq 0.93, 1.03, 1.08) and therefore decline or stagnate. Re-initialised at the observed 1985 stock, keeping each model's own E and τ, the three reach +0.381, +0.412 and +0.379 against an observed +0.309, and SP1 lands on 75.4 exactly. For this family the initialisation constraint (the ratio prior) is the operative fix and the rate anchor is irrelevant — Yasso's rates are fixed by design.

Note however that the Yasso response times (17–38 yr) are *faster* than the 44–87 yr the observations imply, so correcting the initialisation alone would leave them over-accumulating by 23–33 %: they would swing from reporting a source to over-reporting a sink. That is a large improvement on a sign error, and the counterfactual holds E fixed whereas a real recalibration would adjust it — but "necessary and sufficient" overstates it. Necessary, and sufficient for the sign and the order of magnitude.

TP2/TP3: a turnover problem. At $\tau = 195$ yr they close only 18 % of the gap and reach 70.8 even when perfectly initialised. No initialisation fixes them. They require α_H some 2.3–4.7× faster than the ICBM anchor, and — because faster turnover also lowers the equilibrium — the increase must be accompanied by a compensating rise in σ_{input} so that the level is held while the response speeds up.

> **PROPOSAL — FREE SLOW RATE — relax the slow-rate anchor for the simple models**
> Widen the C1 slow-rate prior (currently log SD 0.15, centred on Ultuna arable k_2) so the data can select the turnover time. Report the displacement as the transferability result.
>
> *Why this is not a retreat from C1.* C1's stated design makes the slow-rate posterior width "the arable→forest transferability diagnostic". It has fired: all three simple models sit 1.6–2.2σ from the anchor, consistently slower, and the trend analysis says the required correction is 2.3–4.7×. The *fast* rate stays hard-fixed — it is litterbag-grounded and transferable; only the component whose provenance was flagged as weak (bare fallow, no fresh input, most recalcitrant fraction remaining) is freed.
>
> *It corrects an asymmetry rather than creating one.* The homogenisation principle is equal external *information.* Yasso's rates are fixed because they were calibrated on forest data; the simple models' were anchored to arable data. Those are not of equal quality for a forest application, and freeing the latter restores the intended parity.
>
> *Independent corroboration.* Yasso's forest-calibrated rates give effective $\tau = 23$–38 yr, against 195 yr for the ICBM-anchored cascades. Two independent sources — Yasso's own calibration and the observed Finnish SOC trend — agree that forest turnover is several times faster than the Ultuna arable value.

> These response times are fitted from three trajectory points using the 2084 projection as an equilibrium proxy, on cross-plot means. Directionally solid; the magnitudes need an analytical treatment before publication. And the observed target is itself uncertain: on the two campaigns validated against official LUKE figures (2006→2024) the observed rate is +0.209, not +0.309, so part of the apparent deficit may be the 1985 stock reading low — the C5 concern resurfacing on the trend rather than the level.

8.3 Can the simple structures fit at all?

The response-time analysis appears to condemn TP2/TP3: at $\tau = 195$ yr they close only 18 % of the gap and cannot reach the observed accumulation at any initial state. But that conclusion depends on which parameters are held fixed. TP2 has *three* relevant free knobs, not one:

$$E = J_{\text{eff}} \left[\frac{1}{\alpha_A \xi} + \frac{p_H}{\alpha_H \xi} \right], \qquad \tau = \frac{1}{\alpha_H \xi},$$

so α_H sets the response time, p_H raises the equilibrium *without* slowing the response, and σ_{input} scales both. Faster turnover alone lowers E proportionally and is self-defeating; faster turnover *with* compensating p_H or σ_{input} is not.

Table 6: TP2 feasibility at $\tau = 45\,\text{yr}$ (the fast end, best matching the observed deceleration), for the equilibrium $E = 85.5$ implied by $61.5 \to 75.4$ over 39 yr. NPP ceiling $8.7\,\text{tC}\,\text{ha}^{-1}\,\text{yr}^{-1}$; p_H displacement is against the Tier-2 prior (centre 0.13, logit SD 0.4).

p_H	σ_{input}	flux	p_H displacement	
0.13 (ICBM)	4.81	11.94	$+0.0\sigma$	**exceeds NPP**
0.20	3.34	8.29	$+1.3\sigma$	ok, tight
0.30	**2.33**	**5.77**	$+2.6\sigma$	**comfortable**
0.45	1.60	3.97	$+4.3\sigma$	very comfortable

So the simple structures can fit. TP2 reaches the observed level *and* trend with $\alpha_H \approx 0.025$, $p_H \approx 0.30$ and $\sigma_{\text{input}} \approx 2.3$ — an effective flux of 5.8, comfortably inside the envelope. The apparent impossibility was an artefact of holding p_H at its current value while varying α_H.

What it costs relative to the arable analogue. Humus turnover $\approx 3.9\times$ faster than Ultuna k_2; humification efficiency $0.13 \to 0.30$, a 2.6σ prior displacement but squarely inside the range usually quoted for humification efficiencies (0.1–0.3); and litter input $2.3\times$ the measured product. Each is individually defensible; together they are a substantial departure from ICBM, and the departure is the interesting part — particularly for p_H, where a forest organic layer humifying more efficiently than arable bare fallow is close to expected.

Consequence for the intercomparison. This inverts the reading of C1. The anchor was not making the comparison fair; it was preventing the simple models from reaching the region of parameter space that fits, and the $10\times$ response-time gap against Yasso was the symptom. With forest-appropriate kinetics the two-pool cascade *can* represent Finnish SOC dynamics, which is a positive answer to the question the simple models were included to ask.

> This is a steady-state feasibility calculation on cross-plot means. It establishes that a solution *exists* in parameter space — not that the sampler finds it, nor that it holds plot-by-plot. The three knobs trade off along a ridge, so the calibration may land anywhere along it; watch α_H, p_H and σ_{input} *together*, not α_H alone.

8.4 Correction: the trend is driven by rising litter, not by historical depletion

Sections 8–8.3 analysed the trend as relaxation toward a *fixed* equilibrium, and concluded that the models must start far below it — with a required $R \approx 0.60$, well under the growing-stock value, motivating a historical-depletion hypothesis (litter raking, forest grazing, slash-and-burn). **That analysis was wrong and the conclusion is withdrawn.**

The equilibrium is not fixed: national mean litter rises $\times 1.35$ over 1986–2024. A soil starting *at* equilibrium and simply tracking that rise produces:

Table 7: SOC in 2024 for a soil at equilibrium with 1985 litter, tracking the observed litter rise with response time τ. Observed: 75.4, trend +0.312.

τ (yr)	SOC 2024	trend
23 (Yasso20)	76.6	+0.387
30	74.8	+0.341
45	71.9	+0.267
195 (TP2)	64.6	+0.081

$\tau \approx$ 33–38 yr reproduces the observed accumulation exactly, with **no below-equilibrium start required**. The diagnostic error was the same one that gave SP1 a fitted τ of 79 yr against its true value of 33: a three-point exponential fit assumes a stationary target, and the target is not stationary.

Consequences.

- The **historical-depletion hypothesis is not needed**. It may still be true, but the SOC record does not demand it, and no claim should rest on it.
- The **conflict over R dissolves**. There is no competing trend requirement pulling R below the growing-stock value, so $R \approx 0.90$ (section 5) stands uncontested and can be applied to all models without harming the cascades.
- The **two failure modes are now cleanly separated**, one per family and sharing no parameter: Yasso is initialised *above* equilibrium ($R = 1.22$, 1.51) and relaxes downward — fixed by R; TP2/TP3 are too slow to track the rise ($\tau = 195$) — fixed by $\alpha_H \approx 0.032$, roughly 4.9× ICBM, and unrelated to R.

The mechanism, stated plainly. Finnish forest soils are accumulating carbon because litter input is rising with the growing stock. A model reproduces that only if its turnover is fast enough to track the rise, and only if it is not initialised above its own equilibrium. Both observed failure modes follow from that one sentence.

8.5 σ_{init} is conditionally identified, and the condition is the 1985 campaign

It would be wrong to describe σ_{init} as unidentified. Pushing the prior through the `flux_pair` transform and comparing with the posteriors:

Table 8: Implied prior on σ_{init} versus the fitted posteriors (`20260805` run).

	median	95 % width	narrowing
prior (implied)	**1.000**	**1.950**	—
SP1	0.627	0.322	6.1×
TP2	0.564	0.325	6.0×
TP3	0.566	0.332	5.9×
Yasso07	0.768	0.273	7.1×
Yasso15	1.002	0.397	4.9×
Yasso20	1.236	0.895	2.2×

The data narrows σ_{init} by 2–7× and moves the median by up to 44 % (prior 1.00 → TP2 0.564). Four of the six models pull it a long way down. This is a parameter the calibration genuinely informs.

But the information sits almost entirely in one campaign. 1985 is the observation closest to the initial state, so it carries most of the signal about it. The C5 sweep is the direct demonstration: changing how much the 1985 campaign is trusted moved σ_{init} by a factor of *four* while trusted-campaign RMSE moved 0.45 %. The parameter is well determined *given* a choice about 1985, and that choice is not itself determined by the data.

This is a scope statement, not a defect. The correct form of the inference is conditional: *given that the VMI8 campaign is taken at face value, the reconstructed initial state is X*; down-weighting VMI8 shifts it by up to 4× at negligible cost in predictive skill. Stating it that way is honest and it is exactly the sensitivity the C5 record was kept to support. It should not be presented as an unconditional estimate.

It also *localises* the initialisation problem usefully. The difficulty is not that soil carbon data cannot in principle constrain an initial state — it demonstrably can, by a factor of 2–7. It is that the constraint rests on the one campaign whose layer protocol differs (VMI8 samples 0–5 and 5–20 cm; the later campaigns 0–10 and 10–20 cm) and which could not be validated against official LUKE figures. That is a specific, addressable data question rather than a generic identifiability lament.

And it explains Yasso20. It is the *least* constrained of the six — 2.2× narrowing against 6–7× elsewhere, and a posterior nearly three times wider. So Yasso20 does not merely land in the wrong place; it is the model the data pins down least, which is why it drifts furthest and why the prior matters most there. That is a sharper diagnosis of the sink-sign failure than "σ_{init} is too high".

8.6 Acceptance criteria agreed for the next run

Yasso, effective flux (F9)	must stay well inside the NPP envelope — non-negotiable
SP1/TP2/TP3, effective flux	high flux tolerable; report as a finding
Trend, all models	"reasonably close": within roughly a third of observed
Level, all models	aligned at 1985 (achieved by the likelihood fix)

The two families need qualitatively different fixes, and only one strains plausibility: **Yasso's correction is flux-neutral** (the R prior moves σ_{init}, not σ_{input}, so its contemporary flux of 2.4–3.5 is untouched), whereas **the cascades' correction costs input** ($\sigma_{\text{input}} \approx 2.3$). For a study asking whether simpler structures suffice, that asymmetry is itself a result.

Which observed trend to aim at. Paired rates are $+0.458 \pm 0.048$ (1985–2006), $+0.209 \pm 0.065$ (2006–2024) and $+0.312 \pm 0.035$ (1985–2024). The observed deceleration factor of 2.19 is steeper than pure approach-to-equilibrium predicts (1.18–1.55), which would be the signature of 1985 reading low — but the two rates differ by only $\approx 2\sigma$, so the evidence is weak. We adopt $+0.312 \pm 0.035$ as the primary target (best determined, spans the whole record) and report $+0.209$ as the trusted-campaign sensitivity.

9 Verification before any recalibration

1. Transform round-trip: `to_unconstrained(to_original(x)) == x` for the new pair.
2. Jacobian checked numerically against a finite-difference determinant.
3. `preflight_prior_pushforward.R` at full N: forward blow-up rate should remain $\approx 0\,\%$, as it did on the current target. This gate caught the `beta2` detonation and is the natural one here.
4. Prior pushforward on J_{1917} itself: confirm the implied 1917 flux distribution sits inside the NPP envelope without the guard firing appreciably.
5. ESS on R after the run — a parameter against a boundary mixes worse, the caveat that applied to σ_{input} under `flux_pair`.

10 Plan for the next calibration

Established — goes in regardless

1. **Log-normal likelihood.** Confirmed on TP2: bias $+12.7 \to -0.9$ tC ha^{-1}, median log-residual $-0.205 \to -0.023$. Implemented as `HIKET_LOGNORMAL_LIK`, default off. This is the single largest correction and it is not speculative.

2. **the common anchor — common pre-run anchor.** Reference both ends to $\bar{J}_{t_0}$ so the sampled ratio *is* the 1917/1985 ratio. A defect correction; neutral for fit.

3. **plain priors — collapse the auxiliary priors** to two ordinary lognormals. Paired pre-flight shows identical blow-up rates with and without `flux_pair` in all six models, and 0 % prior mass above the NPP ceiling.

Aimed at the trend — family-specific

4. **the ratio prior — growing-stock prior on** R, for the **Yasso family**. Necessary and sufficient there: their τ is already right and only the starting position is wrong. Expected to convert the reported source back into a sink.

5. **the free slow rate — relax the slow-rate anchor**, for **TP2/TP3**. The only lever on their trend; irrelevant to Yasso, whose rates are fixed. *Under test.*

Still open

6. σ_{input}**'s centre.** C4b set it at 1.30 before the SOC target was corrected downward by a comparable factor. Deliberately not bundled: changing it alongside the error model would make the run uninterpretable. Note the log-normal refit left σ_{input} essentially unmoved, so this may matter less than it appeared.

7. σ_{obs} **fixed** (D4). A correctness issue independent of everything above, and partly superseded: under a log-normal the width no longer biases the location.

8. **Whether the NPP envelope should apply to the 1917 flux at all** — a modern physiological ceiling extrapolated to a forest for which no NPP measurements exist.

9. **The observed target itself.** On the two LUKE-validated campaigns the observed trend is +0.209, not +0.309. Resolve before tuning models to close a gap that may be a third smaller than it looks.

Verification gate before launch

Transform round-trip; Jacobian against finite differences; `preflight_prior_pushforward.R` at full N (≈ 0 % blow-ups expected); prior pushforward on J_{1917} itself; and ESS on R after the run.

Pending at the time of writing

Two short local runs were in progress: TP2 with log-normal + widened slow-rate prior (predicting $\alpha_H \to 0.012$–0.025 and the trend from +0.116 toward +0.24), and Yasso20 with log-normal alone (testing whether the level fix generalises to a fixed-rate model, and whether the inverted sink survives a correctly specified likelihood). Results to be appended.